

Technology Validation and Start Up Fund

Round 34 Proposal Evaluations

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TECHNOLOGY VALIDATION AND STARTUP FUND

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1) Executive Summary

Redwood is a Columbus, Ohio based LLC founded by former Battelle executives over 10 years ago. Redwood has assembled an extraordinary team for this Program. Each member of the five-person Redwood team is an accomplished technology commercialization professional with decades of experience in performing business and technical evaluations. This team, combined with identified external subject matter experts, has extensive experience in all six of the Ohio Third Frontier technology focus areas. More detail on the Redwood team is provided in Appendix 1 of this report and on our website (www.Redwdinnnov.com). Details of the TVSF program and the review process are provided in Appendix 2.

Ten (10) TVSF Round 34 applications, three (3) Phase 1 and seven (7) Phase 2, were received and initially reviewed. Three (3) Phase 1 were withdrawn without further review because they did not meet the TVSF application criteria and one (1) Phase 2 was withdrawn without further review because it was submitted by an ineligible applicant. The four withdrawn applications are not included in the analytic results.

The reviewed proposals consisted of 6 Phase 2 applications totaling \$800,000.

Funding is recommended for two Phase 2 applications for a total of \$250,000. This translates to a 33% recommended application funding rate for this TVSF round, compared to the average of 49% over all 34 TVSF rounds.

2) Evaluation Results

Summaries of the evaluations of the proposals for Round 34, incorporating the interview, and funding recommendations are shown below in Table 1. Rounds 22-29 interviews were conducted via video due to COVID-19 concerns. Round 30 interviews were conducted in person and Rounds 31-34 were conducted via video. The total recommended funding for Phase 2 projects is \$250,000. Note that the Table 1 column widths are proportional to the weighting of the evaluation criteria. For example, in Table 1, Management Team which is weighted at 20 is four times as wide as Growth Plan in Ohio which is weighted at 5. Note that a yellow evaluation indicates that the proposal meets that particular criterion.

More detailed evaluations and recommendations for each Phase 2 proposal may be found in Section 3 of this report.

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Table 1 – Phase 2 Proposal Evaluation and Funding Recommendation

Table 1
Phase 2 Proposal Evaluation and Funding Recommendation
TVSF Round 34

Proposal Number	Lead Applicant	Requested Funding (\$1,000)	Recommended Funding (\$1,000)	Team	Opportunity/Market Size	IP Protection	Prod of Concept	Potential Investor & Business Partner Engagement	Business Model	Project Plan/Budget Narrative	Growth Plan in Ohio	ESP Interaction
FY24-6154	Digital Story Therapies, Inc.	100	100									
FY24-6158	Rescue Ventilation Systems	150	150									
	Sub-Total	250	250									
FY24-6152	Advanced & Innovative Multifunctional Materials, LLC	100	0									
FY24-6155	Koazie Investment Holdings, LLC	150	0									
FY24-6156	Noura Health Launch	150	0									
FY24-6157	OSMI Health, LLC	150	0									
	Sub-Total	550	0	Column width is proportional to score weighting in each category								
	Total	800	250									
Evaluation Scale				Weak	Meets	Exceeds						

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Table 2 lists the funding approval rate by TVSF round. This round's approval rate of 33% of the total submitted proposals is on the low side of the average over all thirty-four rounds. The historical range of recommendations for individual rounds has spanned 27 – 100%, with an average of 49%.

Table 2. TVSF Approval Rate by Round

TVSF Round 34 Approval Rate by Round Phase 2

Round	\$ Recommended	Approval Rate
1 (APR 2012)	\$950,000	35%
2 (AUG 2012)	\$900,000	52%
3 (DEC 2012)	\$610,000	44%
4 (JUN 2013)	\$864,000	30%
5 (FEB 2014)	\$1,462,000	46%
6 (JUN 2014)	\$998,000	39%
7 (OCT 2014)	\$1,100,000	57%
8 (FEB 2015)	\$710,000	37%
9 (JUN 2015)	\$550,000	31%
10 (DEC 2015)	\$925,000	38%
11 (APR 2016)	\$1,239,000	46%
12 (OCT 2016)	\$3,537,269	46%
13 (MAR2017)	\$1,567,500	38%
14 (SEP 2017)	\$498,832	27%
15 (DEC 2017)	\$2,250,000	38%
16 (MAR 2018)	\$2,098,600	52%
17 (SEP 2018)	\$2,100,000	42%
18 (DEC 2018)	\$1,150,000	35%
19 (APR 2019)	\$2,250,000	43%

Round	\$ Recommended	Approval Rate
20 (NOV 2019)	\$1,350,000	43%
21 (FEB 2020)	\$3,944,000	56%
22 (JUN 2020)	\$1,398,630	53%
23 (DEC 2020)	\$900,000	50%
24 (MAR 2021)	\$2,092,900	55%
25 (JUN 2021)	\$800,000	75%
26 (OCT 2021)	\$1,700,000	55%
27 (FEB 2022)	\$850,000	43%
28 (APR 2022)	\$2,499,976	64%
29 (JULY 2022)	\$850,000	100%
30 (OCT 2022)	\$3,700,000	71%
31 (JAN 2023)	\$100,000	50%
32 (APR 2023)	\$850,000	64%
33 (JULY 2023)	\$1,100,000	73%
34 (OCT 2023)	\$250,000	33%
Total Funding	\$48,145,707	
Average/Round	\$1,416,050	49%

3) Proposal Summaries

Proposal Summaries - Phase 2 Recommended for Funding

Proposal 24-6154	Digital Story Therapies, Inc.	Amount Requested: \$100,000
<i>Licensing Institution</i>	Research Institute at Nationwide Children's Hospital	Amount Recommended: \$100,000
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	Hear Me Read

Company Snapshot: Digital Story Therapies' ("DST") digital storybook application utilizes evidence-based speech therapy methods to accelerate reading, speech, and language results for hearing-impaired children.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
G	Management Team	The team has a strong, appropriate mix of backgrounds and credentials. The commitment of the team was evident during the interview.
Y	Opportunity/Market Size	There is a large US pediatric market - which is quoted at \$1.4B. Launch market is the 7 children's hospitals in Ohio. The yellow rating reflects the initial target market.
Y	Intellectual Property Protection	Patent application was filed on 7/1/21 and has been published. PCT initial search was published 12/29/22. Licensing discussions are underway with NCH.
G	Proof of Concept	MVP will be completed in September 2023 with company funding. TVSF funds will enable complementary web capabilities for patient access and payment.
Y	Potential Investor/ Business Partner Engagement	Have secured numerous non-dilutive grants to date. Anticipate next funding after founders' bridge will be from Angels and / or Rev 1.
G	Business Model	Target customer sequencing is well thought out. Revenue model is well explained and is significant before FDA approval.
G	Project Plan/ Budget Narrative	Plan is well explained and appropriate. Company is covering \$11.4K
G	Growth Plan in Ohio	Plan to complete development and hire in Ohio is explicitly provided.
G	ESP Interaction	Extensive involvement with Rev1 is described.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: This is a strong proposal featuring a team that is well positioned to serve a significant unmet market of providing increased access to speech therapy for pediatric deaf and hard of hearing patients.

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Proposal 24-6158	Rescue Ventilation Solutions, LLC	Amount Requested: \$150,000
<i>Licensing Institution</i>	University of Cincinnati	Amount Recommended: \$150,000
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: 33	SPIRITUS

Company Snapshot: Rescue Ventilation Solutions, LLC is developing SPIRITUS, a compact, intuitive, inexpensive, self-powered, disposable device for use on patients requiring manual ventilation with a bag-valve masks (BVM) in the prehospital or hospital setting.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
Y	Management Team	Management team has commercial, technical and clinical experience and includes the inventors of the device. Staffing will be increased as needed to support future business activities.
Y	Opportunity/Market Size	Proposal makes the case for a low cost, reliable, and easy to operate device. Potential market is large. Possible pricing constraint in EMS market will be tested with higher margin markets/offers.
Y	Intellectual Property Protection	Pending patent covers device and method of use. It is not apparent that significant trade secret / know-how will appreciably enhance the competitive barriers to entry.
Y	Proof of Concept	Current prototype TRL 5/6 (demo in relevant environment) with lower cost sensor identified. Project team is well positioned to know market needs and performance requirements of the proposed device.
Y	Potential Investor/ Business Partner Engagement	Previous UC Venture Lab/TVSF Phase 1 funding. Relationships with local VC firms. Strong contacts within the medical community.
Y	Business Model	Initial focus on medical provider market using direct/distributor sales channel. Pro forma Revenue \$14.4 MM/GM 63%/NM 45% (Year 5).
Y	Project Plan/ Budget Narrative	Project plan/contractors/pricing provided for proposed project work.
Y	Growth Plan in Ohio	Proposal confirms plans to remain and hire in Ohio.
Y	ESP Interaction	ESP has participated in preparation and supports proposal.
	Evaluator Recommendation	This application is recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Balanced management team with relevant commercial, technical, and clinical expertise. Initial market strategy focused on direct/distributor sales channels to medical providers (EMS/hospitals). Pricing constraints (reimbursements) in EMS setting are challenging from an investor perspective but value pricing that recognize improved patient outcomes will be tested. Longer term, other market segments, product offerings, and delivery strategies with the potential to increase revenue and profit margins will be pursued.

Proposal Summaries - Phase 2 Not Recommended for Funding

Proposal 24-6152	Advanced & Innovative Multifunctional Materials LLC	Amount Requested: \$100,000
<i>Licensing Institution</i>	University of Dayton	Amount Recommended: \$0
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: 32	<i>Short Chain PFAS Adsorption via AimmSorb</i>

Company Snapshot: Advanced & Innovative Multifunctional Materials LLC developed a platform technology that enables standard carbon-based porous materials to possess tunable properties. This application is tailored to adsorb, and remove Per- and polyfluoroalkyl substances (PFAS).

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	A very strong Team with technical/scientific experience and track record. However, seriously lacks business development, commercialization, marketing and scale up skill sets to deliver projected financial outcomes.
G	Opportunity/Market Size	A very large market in B dollars/ yr. growing at > 6 % CAGR.
Y	Intellectual Property Protection	Filed for a patent application and PCT search report looks encouraging. Exclusive license is in place.
Y	Proof of Concept	Prior Phase I work shows good technical feasibility.
Y	Potential Investor/ Business Partner Engagement	Some very early discussions. Waiting for further data/ validation for the value proposition.
R	Business Model	Solely based on licensing of the technology. Projected revenues and licensing path seem very optimistic.
Y	Project Plan/ Budget Narrative	A well laid out project plan with milestones.
Y	Growth Plan in Ohio	Plans are to grow the business in Ohio.
Y	ESP Interaction	Working actively with local ESP.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: The Team is technically very strong and credible. Early results at 1 to 2-gram scale shows potential in mitigating short chain PFAS with favorable recyclability. Contacts with potential licensees with early lab data has generated interest. The Team seriously lacks the needed business development, commercialization, licensing, and scale up skills to realize the projected business revenues, exclusively through licensing of the technology. Market opportunity is very large and a practical solution to address the PFAS mitigation is real and needed. The Team is encouraged to resubmit the proposal addressing the deficiencies noted.

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Proposal 23-6155	Koazie Investment Holdings LLC	Amount Requested: \$150,000
<i>Licensing Institution</i>	Bon Secours Mercy Health Foundation	Amount Recommended: \$0
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	<i>Infection Prevention and Compliance Survey Tool</i>

Company Snapshot: Koazie Investment Holdings LLC is developing a cloud-based mobile application that equips users in healthcare settings with a one-handed tool for recording compliance surveys.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	Identity of management team was not provided in the proposal. Corporate governance and structure remain unclear and confusing. Proposed CEO is key to success but affiliation, expertise and track record are not well described.
Y	Opportunity/Market Size	US Market is 15.6K nursing homes and 6K hospitals. With different pricing strategies for these different market segments, estimate market potential for the initial product to be over \$200 million.
Y	Intellectual Property Protection	Product is software. Copyright and trade secret protections in place. License discussions underway.
G	Proof of Concept	Builds on Phase 1 work to make prototype version more robust, adaptable to different customers' needs, and building the website to support the app.
Y	Potential Investor/ Business Partner Engagement	Plans to bootstrap fund via product sales, not venture investment. Potential customers interviewed.
R	Business Model	Corporate structure, especially holding company and relationship to subcontractor are confusing. Expertise is not supported.
Y	Project Plan/ Budget Narrative	Project plan and budget achievable within the funding and time.
Y	Growth Plan in Ohio	Commitment to grow the business in Ohio.
Y	ESP Interaction	Engaged with Entrepreneur in Residence at Mercy Health.
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Interview described a holding company that houses three related software products, one of which is the subject of this application. The relationship between the holding company, this product and the subcontractor are not well defined. Subcontractor appears to be part of the company and vice-versa. CEO would be the key individual responsible for company/product success; however, background and expertise are not well described. This proposal is not recommended for funding.

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Proposal 24-6156	Noura Health, Inc.	Amount Requested: \$150,000
<i>Licensing Institution</i>	Bon Secours Mercy Health Foundation	Amount Recommended: \$0
Prior Phase 1 Applications: Yes	Prior Phase 2 Applications: No	Noura Health Launch

Company Snapshot: Noura Health is a personalized nutrition plan that works as a whole-body approach to wellness.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	The management team consists of 2 individuals who have recently left senior management positions and formed team. Roles evolving. They have broad skills in marketing, digital consumer and healthcare markets. Two advisors have fund raising experience.
Y	Opportunity/Market Size	Direct-to-Consumer lab testing is identified as the Total Addressable Market at US \$2.4 B in 2020. The service addressable market is quantified at \$1B.
Y	Intellectual Property Protection	Software components are copyrighted and will be included in the license to Noura Health. Trademark has been filed.
G	Proof of Concept	TRL is currently at 7 and developed protocol with over 15,000 patients. Expect an increase to 8 as the MVP is completed by mid-October 2023.
R	Potential Investor/ Business Partner Engagement	Team raising \$1.3 M by end 2023. Investors may be patients. Total investment of \$13.3 M anticipated. No potential partners identified.
Y	Business Model	The business model assumptions include nutrition analysis via proprietary algorithms. 2027 net profit is targeted at over \$61M.
Y	Project Plan/ Budget Narrative	Narrative is consistent with the project plan.
Y	Growth Plan in Ohio	The plan is to grow the company primarily in Ohio.
Y	ESP Interaction	Interaction has been through EIR. JumpStart did market & user work
	Evaluator Recommendation	This application is not recommended for funding.

Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: There is significant research provided by a clinician to hone the algorithms that will be used in the nutritional analysis. The company is using software in a medical application. No potential partner interactions identified or considered. A raise of \$1.3 M by years end is unrealistic as it has yet to be presented to potential investors. Questions relating to pro forma changed the net profit from \$24M to over \$61M. The team, while seasoned, is newly formed and roles are evolving. |

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Proposal 24-6157	OSMI Health, LLC	Amount Requested: \$150,000
<i>Licensing Institution</i>	Ohio State University	Amount Recommended: \$0
Prior Phase 1 Applications: No	Prior Phase 2 Applications: No	COVID-19 Breath Test

Company Snapshot: OSMI Health, LLC is developing a handheld and portable breathalyzer device utilizing a resistive catalytic sensor to detect COVID-19 validated in hospitalized population and in an ambulatory study.

Rating (R/Y/G)	Category	Highlights/Issues/Comments
R	Management Team	No specific medical device experience, technical or business development, on the team. Recently recruited remote CEO. Team makes overoptimistic assumptions about readiness of the product that are not substantiated in the proposal.
Y	Opportunity/Market Size	The COVID testing market is \$24-25B in 2023, and even though it's expected to decline ~11-13% per year, the market is forecast to be very large for many years.
Y	Intellectual Property Protection	A patent application exists for the subject product. Company will enter negotiations for license.
Y	Proof of Concept	Product prototypes were previously tested, and performed well enough to deserve further development. The applicant seeks to test performance in two other geographic locations, per FDA contacts.
R	Potential Investor/ Business Partner Engagement	Brief initial discussions with VC firms appear theoretical rather than substantive. Funded separately by DTRA but refused by BARDA.
R	Business Model	Proforma is not credible. Many aspects of company formation and management are lacking.
Y	Project Plan/ Budget Narrative	Project plans within budget and timeframe proposed.
R	Growth Plan in Ohio	CEO in CA. No concrete plans for personnel or facilities in Ohio.
Y	ESP Interaction	Obtained help from Rev1.
	Evaluator Recommendation	This application is not recommended for funding.

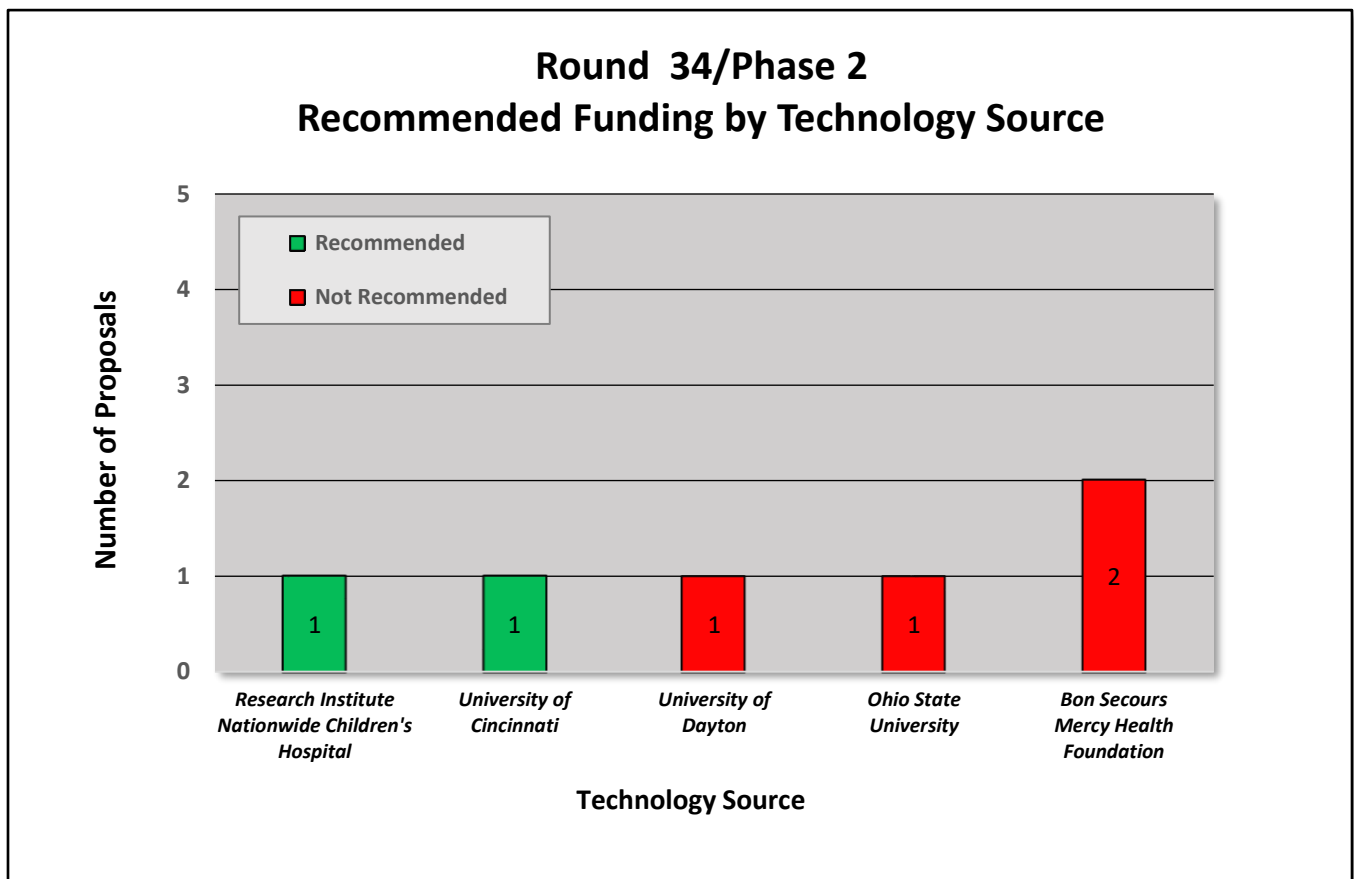
Evaluation Scale	Weak	Meets	Exceeds
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Comments and Recommendations: Medical device development experience is lacking. Proposal overstates readiness of the product and underestimates what it will take to complete product development. Market readiness not substantiated in the proposal. Interview indicated more data may be available but was not available to the reviewers. Many aspects of diagnostic device development are missing, raising doubts about of revenues forecast in 2024. This application is not recommended for funding.

4) Round 34 Analysis

Figure 1 shows the proposal activity and funding recommendations by technology source for Phase 2 proposals. Bon Secours Mercy Health Foundation was the most active technology source with two applicant submissions. One applicant submission using technology sourced from the Research Institute Nationwide Children's Hospital and one from University of Cincinnati are recommended for funding.

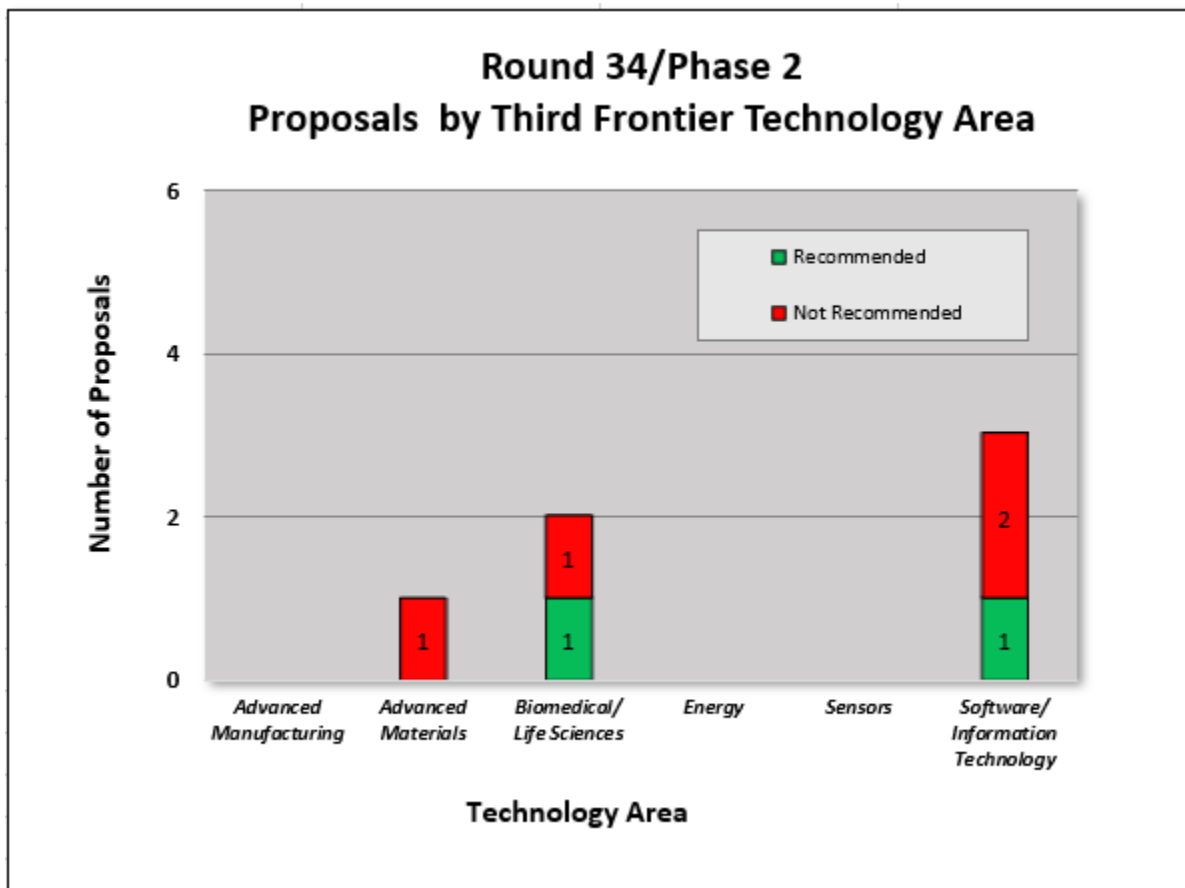
Figure 1. Round 34 Funding by Technology Source



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Figure 2 depicts Phase 2 proposal activity and funding recommendations by Third Frontier focus area. In this Round, three of the six proposals were in Software/Information Technology (50%), two of six were in Biomedical/Life Sciences (33%), one of the six proposals was in Advanced Materials (17%). One of six Biomedical/Life Sciences (17%) and one of the six in Software/Information Technology (17%) were recommended for funding. This round represents 67% in Third Frontier Technology areas that were non-Biomedical/Life Sciences. The previous fourteen rounds have averaged 56% of the applications in Biomedical/Life Sciences.

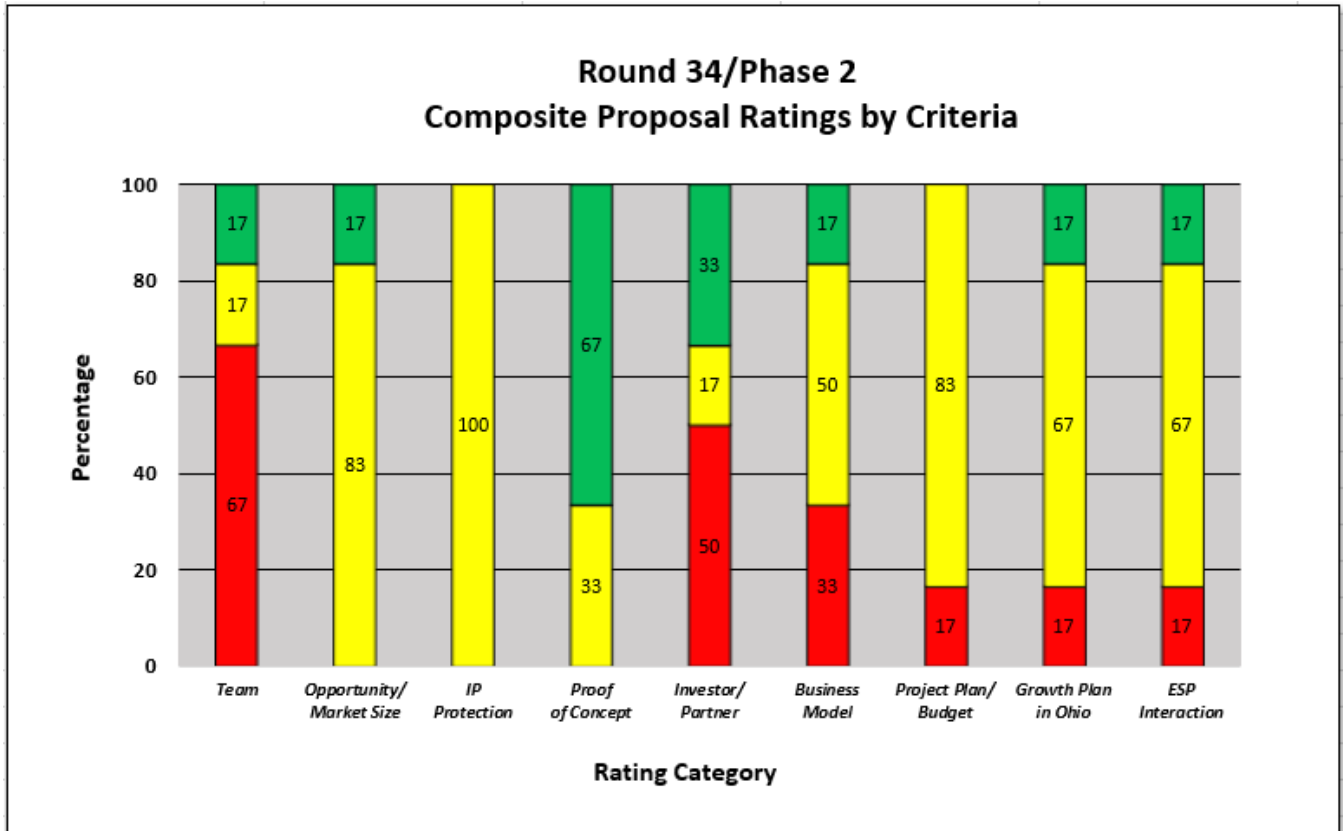
Figure 2. Round 34 Phase 2 Proposal Activity by Third Frontier Technology Area



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Figure 3 shows the aggregate ratings by evaluation criteria for all Phase 2 proposals. Setting aside the Growth Plan in Ohio and ESP interaction categories, the Team, then Potential Investor/ Business Partner Opportunity had the highest percentage of weak scores in this round. IP Protection, Market Opportunity, and Proof of Concept, had the highest percentage of meets or exceeds scores in this round.

Figure 3. Round 34 Phase 2 Proposal Rating Summary



In the previous fourteen Rounds, business model was the lowest rating in Rounds 20-23 (53% average $\geq$ meets) and Round 26 (28%). The RFP was revised prior to Round 24 to elicit stronger business models and it appears that the proposals have provided stronger business models in Rounds 24-34 (64% average $\geq$ meets). Round 26 appears to be an anomaly. Further rounds will be monitored to see if the improvement continues.

Figure 4: Rounds 20 to 34 Phase 2 Analysis of Business Model

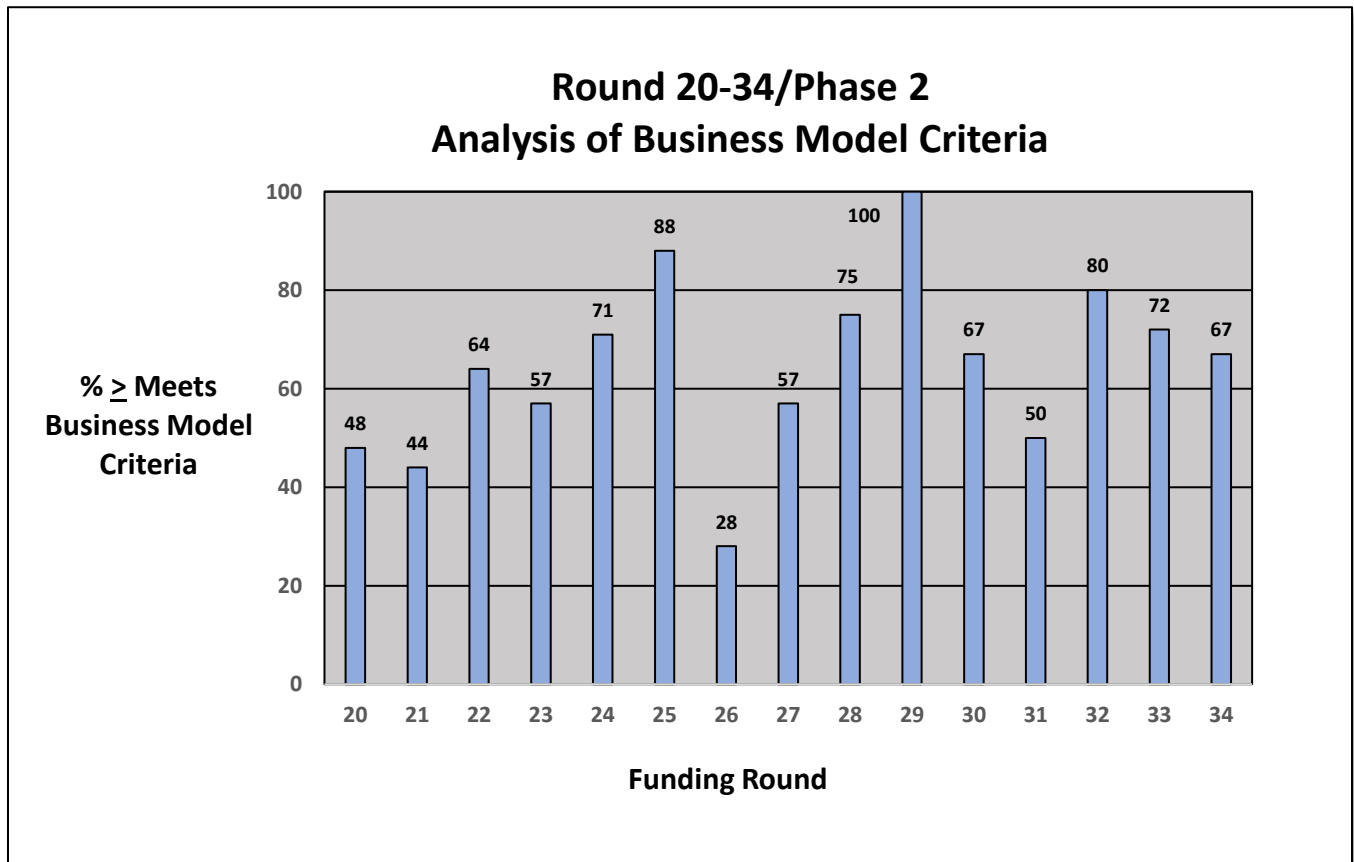
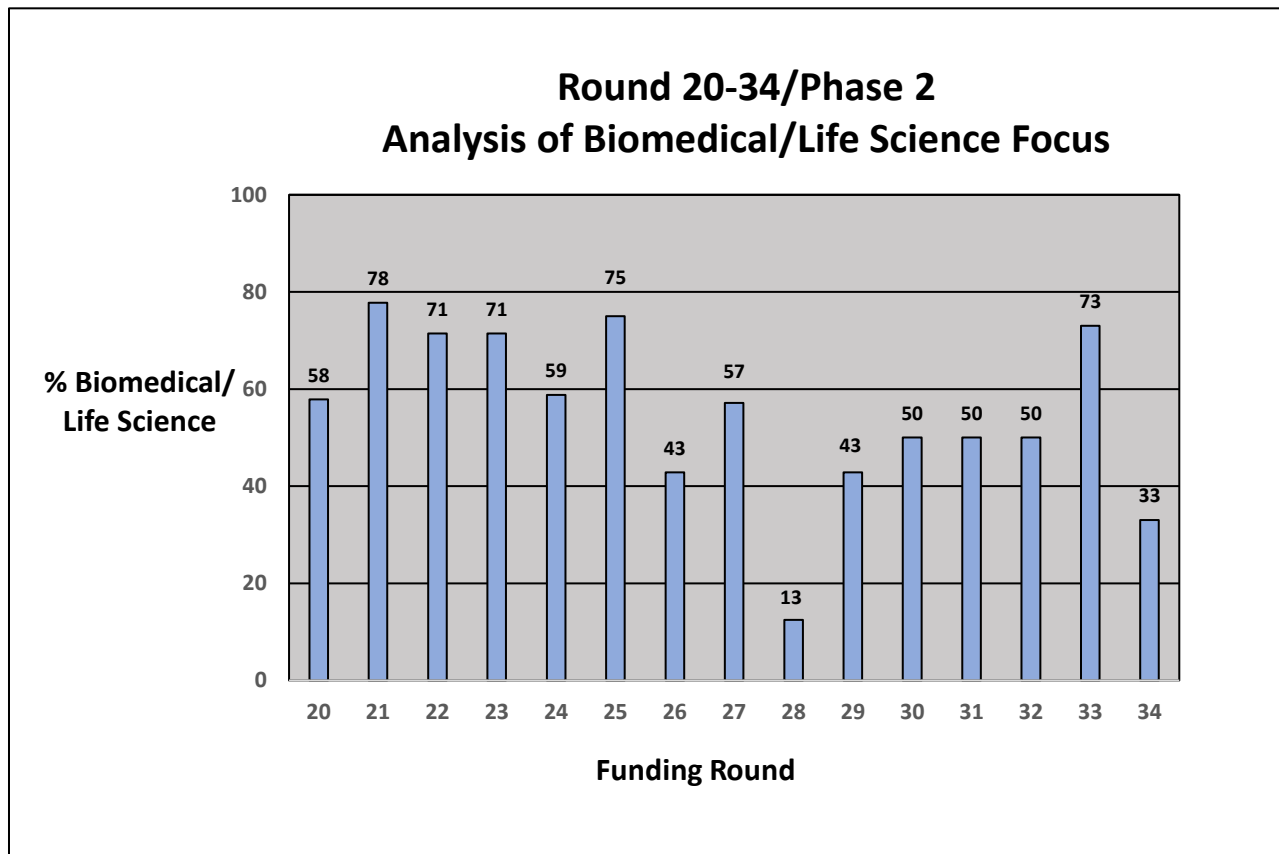


Figure 5 shows the percentage of Biomedical/Life Sciences applications for the last 15 Rounds. Round 34 represents 33% in Third Frontier Technology areas that are Biomedical/Life Sciences. Average of the last 7 rounds is 45%. All fifteen rounds average 56% of the applications in Biomedical/Life Sciences.

Figure 5: Rounds 20-34 Phase 2 Analysis of Biomedical/Life Science Focus



Carry Through and Reapplication

Phase 1 Carry Through: There were 4 Phase 2 applicant that previously received Phase 1 funding. One is recommended for funding.

Phase 2: There were 2 reapplications in this Round. Both were first time reapplications and one is recommended for funding.

Appendix I

Summary of Redwood team and qualifications

Redwood, as a company, has been providing technology commercialization services for over 9 years while each team member has been active in this field for over 25 years.

Each Redwood team member

- possesses an advanced technical degree and extensive business proficiency
- has worked across the spectrum of technology commercialization from invention to successful market introduction
- understands how to assess a concept case from the perspective of aligning technologies to product applications in specific markets
- has lived, both conceptually and literally, the iterative process of understanding market needs and wants, value chains and who the customers are within the value chain

Team members have all worked for major corporations, research institutions, venture capital firms and technology start-up companies gaining a comprehensive understanding of what is necessary for development teams to successfully commercialize a technology. The Redwood team has served as evaluators for the Ohio Advanced Manufacturing program and an individual team member served as an evaluator for CALF, TIP and IOF loan programs for over a decade.

The five members of the Redwood team are highly qualified evaluators for the TVSF program and have combined experience and expertise in the following areas (combined years):

Commercializing technology into market pulled products (125+ years)

Market/Technology Assessment (140+ years)

Startup/ Spin out companies (50+ years)

Board member/Advisor to Startups (30+ years)

Evaluating/ monitoring RFPs/ Funding selection (40+ years)

The following is a brief summary of the five principal team members used in this evaluation Round.

Herb Bresler

- BS Biological Sciences, University of Maryland; BS Secondary Science Education, University of Maryland; PhD Immunology and Infectious Diseases, The Johns Hopkins University School of Hygiene and Public Health
- Former Senior Research Leader and Chief Scientist for Health and Life Sciences, Battelle Memorial Institute, responsible for evaluation of new technology-based business opportunities, intellectual property development, licensing and tech transfer; created and implemented new metrics to increase returns on discretionary R&D; cultivated approximately 1150 invention disclosures, 900 patent applications, and 120 granted patents, leading to \$52 million company funding
- Recipient of four R&D 100 awards for breakthrough medical devices in neuroscience and diagnostics
- Former Director of the Laboratory of Cellular Immunotherapeutics at the Arthur G. James Cancer Hospital and Research Institute at The Ohio State University

John McArdle

- BE, Manhattan College, MS, Northeastern University, Chemical Engineering
- MBA, Finance / International Business, University of Chicago (Booth School of Business)
- Former Business Development Manager, Battelle
- Former Product Line Manager – Koch Industries
- Former Technical Sales Manager, Allied Signal Corporation
- Recognized expert in water and wastewater treatment technologies
- Successful track record of introducing innovative technologies for a variety of municipal, industrial, and military applications in domestic and overseas markets.

Jim Sonnett

- BS, University of Virginia, MS, University of Massachusetts, PhD, University of Delaware, all in chemical engineering
- Former Vice President – Science and Technology, Battelle Health & Life Sciences
- Former R&D Leader – W. L. Gore & Associates and E. I. DuPont
- Built and led high impact innovation organizations in aerospace, electronics, and life sciences
- Former Board Member – Velocys, Ventaira, Battelle Ventures
- Adjunct professor – Ohio State University Fisher School of Business

Susan Stanton

- BS, Millersville University, Chemistry, MPh, Syracuse University, Organic Chemistry, PhD, University of Rochester, Organic Chemistry
- Personally developed 12+ products and led new product development teams at Mobay, Alcoa & Nexicor
- Holder of 10+ patents
- Former VP Market and Technology Assessment at the National Technology Transfer Center

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- Over 15 years as an angel investor in technology-based startups
- Over 15 years as an evaluator for Ohio Third Frontier funds including IOF, CALF and TIP
- Over 8 years teaching market and business analytics to STEM graduate and post doc students.

Bhima Vijayendran

- BS, University of Madras, MS, University of Madras, PhD, University of Southern California in Polymer and Surface Science, MBA, University of New Haven
- Former Senior Research Leader and Vice President Business Development, Battelle Memorial Institute; Chief Research Officer, Battelle Science and Technology, Malaysia
- Former Director, Discovery Research, PPG Industries
- Recognized as one of the leading authorities on advanced materials, special chemical and polymer systems in numerous markets including: Renewable and clean technology, Energy, Nano Technology and Industrial Products.
- Recipient of ten R&D 100 awards and over 100 patents and numerous other awards.

Appendix 2

TVSF objectives and phases

The Technology Validation and Start-up Fund (TVSF) provides grants under two phases to transition technology from Ohio Eligible Research Institutions into the marketplace through Ohio start-up companies. Under Phase 1, Ohio Research Institutions may apply for a pool of funds to support validation/ proof that will directly impact and enhance both the commercial viability of their unlicensed technologies and ability to support a start-up company. Under Phase 2, Ohio start-up and young companies may apply for funding to commercialize a technology they intend to license from a university or an Ohio research institution.

The goals of Phase 1 include:

- Generate the proof needed to move technologies to the point that they are either ready to be licensed by an Ohio start-up company or deemed unfeasible for commercialization. The institutions are encouraged to work with potential Ohio licensees to identify the proof needed.
- Perform validation activities such as demonstration and assessment of critical failure points in subsequent development, prototyping, scale-up and commercialization in order to generate this proof with strong preference for these activities being performed by an independent 3rd party source.

The goals of Phase 2 include:

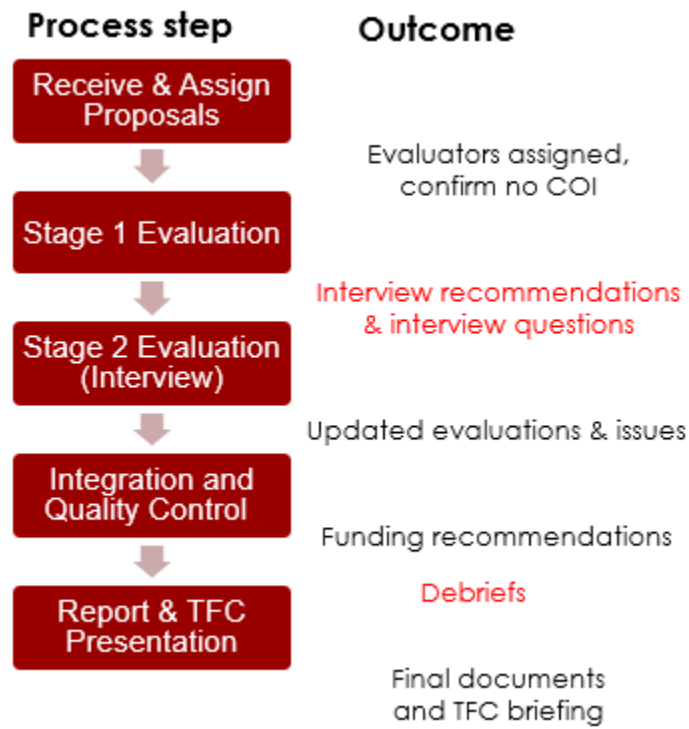
- Accelerate the commercialization of technology by Ohio start-up companies that license technology developed at Eligible Institutions during the critical early stage of life of the company.
- Generate the proof needed to move technology to the point where it is able to be commercialized or additional funds for commercialization can be raised. A clearly identified path to subsequent funding opportunities and working directly with potential investors to define the proof needed for investment into the company is strongly encouraged.
- Funded activities may include, but may not be limited to, beta prototype development and deployment to potential customers for testing and evaluation and market research/ business development in order to generate the proof needed.

Based upon these goals, the proposal evaluation criteria were developed. The proposals were then evaluated based on the criteria.

Description of review process

Review summary. Our overall review process flow and outcomes by stage are shown in Figure 1. A similar process has been successfully used by Redwood in prior projects for public and private clients. Discussions were held with the TVSF program manager after all but the initial step in Figure 1.

Figure 1. TVSF Evaluation Process



Review and Assign Proposal In this first step proposals were summarized and a primary evaluator was assigned who has the appropriate background and no conflict of interest.

Stage 1 Evaluation Stage 1 evaluations were conducted for each proposal using the criteria shown below in Tables 1 and 2. Differentially weighted criteria were used to evaluate Phase 1 and Phase 2 proposals. Each proposal was rated on a 0 (absent) – 5 (Outstanding) scale for each criterion, an approach used by the NSF and in other State of Ohio programs. The weightings reflect the experience of the Redwood team and our belief that some factors, for example team and market opportunity in Phase 2, are more important than others.

The entire review team subsequently discussed all the evaluations to ensure consistency and agreed upon which applicants to invite for interviews. Interview questions were then provided in advance to each applicant.

TECHNOLOGY VALIDATION AND STARTUP FUND

Stage 2 Evaluations (Interviews) The standard procedure for this step is: In-person, 45-minute interviews were held with each invited applicant to discuss the advance questions plus other topics of interest to the evaluators. Two Redwood team members participated in the interviews in person with additional team members joining via Zoom or a conference call.

Integration and Quality Control Proposal evaluations were updated based on interview results. A calibration review was held by the review team to ensure that evaluations were performed consistently and that any changes made were a result of team consensus. Based on this review, proposals were recommended for funding.

Table 1 – Phase 1 Evaluation Criteria

Criterion	Weighting	Description
Alignment and Compliance	Go / No go	Institutional alignment with TVSF intent and compliance with RFP
Project Selection Committee	20	Skills, background and commitment of the committee members
Deal Flow; Budget Strategy	15	Is the projected deal flow consistent with the requested budget to enable committing funds within 1 year?
External Participation	15	Does process ensure validation activities will be performed by 3 rd parties; ESPs and state-funded programs/organizations are enlisted to enhance commercialization activities of the project?
Track Record	15	Is there a strong Phase 1 or comparable program track record of licensing and newco creation? If not, is there a plan for improvement?
Metrics	15	Realism and impact of proposed metrics, including licensing, start-ups.
Project Management & Experience	15	Is there a strong project management strategy and appropriate experience of people who allocate the pool of funds and manage individual projects?
Project Selection Process	5	Is there a clear, appropriate process for project selection?

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Table 2 – Phase 2 Evaluation Criteria

Criterion	Weighting	Description
Alignment & compliance	Go / No Go	Proposal alignment with TVSF intent and compliance with RFP
Management Team	20	Skills, background and commitment
Opportunity / market size	15	What is the market segment and total addressable market? Is it a platform or breakthrough technology or incremental improvement? If breakthrough, is it compatible with viable commercialization pathways?
IP Protection / License	15	Is IP adequately protected, does it enable the business model, is it differentiated from likely competition, is license likely within 9 months?
Compelling Proof of Concept	15	Was meaningful input from potential customers and key performance metrics used to design Proof of Concept? Are the competitive advantages compelling for potential customers?
Potential Investor / Business Partner Engagement	10	Is there company engagement / collaboration independent of licensing institution, including financial backing?
Business Model	10	Is the business model realistic AND achievable? Can the service / manufacturing model be scaled?
Project Plan / Budget Narrative	5	Is the budget consistent with proof in 1 year?
Start-up in Ohio	5	Does a start-up exist or is it planned? Will the start-up be in Ohio?
ESP Interaction	5	Is team engaged with ESP? Has team incorporated feedback from ESP into the project, proposal or business plan?